Choosing an SLM for semantic traffic priority reasoning in SUMO

Qwen3-4B and Phi-4-mini lead the field of 1B–8B parameter SLMs for semantic traffic priority reasoning, but neither is unconditionally superior. Qwen3-4B delivers the highest reasoning scores in its thinking mode and has first-class NVIDIA Jetson tooling, while Phi-4-mini offers the strongest single-mode reasoning with no thinking/non-thinking complexity. Gemma-3-4B-IT posts the highest instruction-following score (**IFEval 90.2%**) of any sub-5B model— LLM Stats +2 critical for structured JSON output—but lacks native function-calling templates. For a simulation-first MSc dissertation, the choice reduces to whether maximal reasoning quality, latency-sensitive operation, or parameter efficiency matters most. No single model dominates all three. Domain-adapted systems like Traffic-R1 and LLMLight provide important benchmark anchors but have narrow evaluation contexts (CityFlow, not SUMO) and limited general reasoning capability. This report evaluates every candidate against an eight-field template, flags all evidence grades, and presents trade-offs across three distinct decision scenarios.

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§1. Methodology and evidence grading

All claims in this report follow a strict source hierarchy. **Tier 1** (peer-reviewed): papers published at KDD, EMNLP, ICML, or equivalent venues—cited as such. **Tier 2** (independent leaderboards): Open LLM Leaderboard, LMSYS Chatbot Arena, HELM, Berkeley Function Calling Leaderboard (BFCL), and llm-stats.com aggregations. **Tier 3** (vendor-reported): model cards, technical reports on arXiv (not yet peer-reviewed), and vendor blog posts—explicitly flagged throughout. **Tier 4** (community): GitHub discussions, Reddit benchmarks, independent blog tests—flagged as "community-reported."

Hardware benchmarks reference the **NVIDIA Jetson Orin Nano 8GB in 25W Super Mode** (1,024 CUDA cores, 102 GB/s LPDDR5 bandwidth, ~5.2 GB usable after OS). Ericxliu ericxliu The primary inference data comes from NVIDIA's official MLC API benchmarks with q4f16_ft quantization (https://developer.nvidia.com/blog/nvidia-jetson-orin-nano-developer-kit-gets-a-super-boost/), which serve as reasonable proxies for llama.cpp Q4_K_M performance. Community tests confirm MLC and Ollama/llama.cpp deliver comparable throughput on this hardware. Where no Orin Nano data exists, extrapolations are flagged. TensorRT-LLM INT4 AWQ benchmarks on Orin Nano are marked "N/A" for all models—NVIDIA's TensorRT Edge-LLM tutorial demonstrates Qwen3-4B running successfully but publishes no throughput numbers. NVIDIA Jetson AI Lab

Reasoning benchmarks are drawn primarily from the Phi-4-mini technical report cross-comparison table (arXiv:2503.01743, vendor-reported by Microsoft), which evaluates multiple models under identical conditions—the most apples-to-apples source available. Qwen3-4B benchmarks come from Alibaba's technical report (arXiv:2505.09388, vendor-reported). Where scores from different evaluation pipelines are available, discrepancies are noted.

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§2. Per-model profiles

Qwen3-4B — the reasoning-quality leader with a latency caveat

(1) 4.0B parameters (arXiv) (3.6B non-embedding), dense decoder-only transformer with GQA, SwiGLU, RoPE, RMSNorm. (Open Laboratory +4) Released **April 28, 2025**. (Artificial Analysis) Architecture family: Qwen3.

(2) Reasoning benchmarks (vendor-reported, arXiv:2505.09388): **MMLU 72.99** (5-shot, base), **GSM8K 87.79** (4-shot CoT, base), **BBH 72.59** (3-shot CoT, base). (arxiv) (arXiv) Instruct thinking mode: IFEval strict **87.75%**, MMLU-Redux 92.77, MATH-500 **95.16%.** (Readthedocs) Non-thinking mode: IFEval strict 68.88%, MMLU-Redux 83.13. (Readthedocs) The base model outperforms Qwen2.5-7B (a model nearly 2× larger) on MMLU, BBH, and GSM8K. (Emergent Mind +2) ARC-Challenge not reported in Qwen3's benchmark suite.

(3) Structured output: native function calling via `<tool_call>` tokens; **0.920 score** on an independent 20-model community tool-calling benchmark (https://github.com/MikeVeerman/tool-calling-benchmark, Round 3, community-reported). (GitHub) Qwen-Agent framework provides canonical function-calling implementation. (Qwen) Supports constrained decoding via llama.cpp GBNF, XGrammar/SGLang, and vLLM guided generation. The updated **Qwen3-4B-Instruct-2507** (July 2025) improves instruction following and tool use further (vendor-reported, HuggingFace model card). (Hugging Face) (GitHub)

(4) Latency on Jetson Orin Nano: **estimated ~32–38 tok/s decode** (extrapolated from similar-size models with published data; no direct Qwen3-4B llama.cpp benchmark on Orin Nano exists). NVIDIA's TensorRT Edge-LLM tutorial uses Qwen3-4B-Instruct as its reference model on Orin Nano 8GB with INT4 AWQ (jetson-ai-lab) (~2 GB weights), confirming deployability but publishing no tok/s figure. (NVIDIA Jetson AI Lab) TTFT estimated **0.8–1.2s** for a ~200-token prompt (extrapolated from Eric Liu's Orin Nano measurements, community-reported, https://ericxliu.me/posts/benchmarking-llms-on-jetson-orin-nano/). TensorRT-LLM INT4 AWQ: N/A—no public Orin-class benchmark.

(5) Memory at Q4_K_M: ~**2.5 GB** (estimated from 4.0B × 0.604 bytes/param + overhead). INT4 AWQ: ~2.0 GB (confirmed by NVIDIA tutorial). (NVIDIA Jetson AI Lab)

(6) Licence: **Apache 2.0**—full commercial use permitted. (DEV Community)

(7) Weaknesses: The thinking/non-thinking mode introduces complexity. (arxiv) (arXiv) Multiple GitHub issues document Qwen3-4B emitting reasoning tokens even with `/no_think` enabled (GitHub) (ollama/ollama#12907, community-reported). The hard switch (`enable_thinking=False` via chat template) is more reliable GitHub but not foolproof. In thinking mode, the model generates hundreds to thousands of reasoning tokens before answering, making it unsuitable for latency-constrained inference. Substack The IFEval++ study (arXiv:2512.14754, independent) found that even top models show **18–61% reliability drops** under prompt perturbation; Qwen3-0.6B showed a 61.8% drop, though Qwen3-4B with rejection sampling can surpass stronger models. arXiv On jailbreak susceptibility, a large-scale study of 63 SLMs (arXiv:2503.06519, independent) found **47.6% of SLMs show high susceptibility** (ASR > 40%); arXiv arxiv Qwen3-4B was not individually reported.

(8) Control-loop evaluation: **No.** Not evaluated in traffic signal control or any safety-critical control loop. Traffic-R1 (§2.11) is built on Qwen2.5-3B, not Qwen3-4B.

Phi-4-mini — strongest single-mode reasoner at 3.8B

(1) 3.8B parameters, dense decoder-only transformer with 200K vocabulary, GQA, shared embeddings. huggingface +2 Released **February 1, 2025**. Architecture family: Phi-4.

(2) Reasoning benchmarks (vendor-reported, arXiv:2503.01743): **MMLU 67.3** (5-shot), **GSM8K 88.6** (8-shot CoT), **BBH 70.4** (0-shot CoT), **MMLU-Pro 52.8** (0-shot CoT), **ARC-C 83.7** (10-shot), TruthfulQA MC2 66.4. huggingface arXiv IFEval: **~63.0%** (vendor, via Prem.ai compilation). Phi-4-mini's BBH score of 70.4 is the highest among all sub-4B general-purpose models evaluated here, indicating strong compositional reasoning. Awesome Agents

(3) Structured output: **native function calling** via `<|tool|>`/`</tool>` special tokens. huggingface Described as "agent-ready, designed with function calling and structured (JSON-style) outputs in mind" BentoML (vendor, HuggingFace model card). Hugging Face Community tool-calling benchmark score: **0.780** (Round 3)—lower than Qwen3-4B's 0.920, with notable variance between rounds (0.880 in Round 2). GitHub Compatible with vLLM Hermes-style tool parser.

(4) Latency on Jetson Orin Nano: **estimated ~35–40 tok/s decode** (extrapolated from Phi-3.5-mini's official 38.1 tok/s, same parameter count and similar architecture). No direct Phi-4-mini benchmark on Orin Nano. TTFT estimated ~0.8–1.0s. TensorRT-LLM INT4 AWQ: N/A—no public Orin-class benchmark.

(5) Memory at Q4_K_M: **~2.3 GB**. The 200K vocabulary slightly increases GGUF size versus comparable 3B models. huggingface

(6) Licence: **MIT**—maximally permissive.

(7) Weaknesses: lower IFEval (~63%) versus Gemma-3-4B (90.2%) and Qwen3-4B thinking mode (87.75%), suggesting weaker adherence to complex formatting instructions without constrained decoding. Weaker multilingual performance than Qwen models (MGSM 63.9 vs Qwen2.5-3B's 53.5, but multilingual MMLU only 49.3 vs 55.9). huggingface arXiv Microsoft's red-teaming covers 23 languages. No thinking mode available—reasoning depth is fixed.

(8) Control-loop evaluation: **No.**

Gemma-3-4B-IT — instruction-following champion

(1) 3.88B parameters, dense transformer with interleaved local-global attention, multimodal (text + vision via SigLIP encoder), 128K context. Hugging Face Released **March 2025**. Architecture family: Gemma 3.

(2) Reasoning benchmarks (vendor-reported, Google via llm-stats.com): **GSM8K 89.2%** (0-shot IT), **IFEval 90.2%** LLM Stats Awesome Agents (0-shot IT— LLM Stats highest of any model under 5B in this survey). LLM Stats Base model from Qwen3 cross-comparison: MMLU 59.51, BBH 51.70, GSM8K 43.97 (base, not IT). arxiv arXiv The IT model's GSM8K of 89.2 is the highest math score among all candidates. ARC-Challenge: not found for 4B IT specifically; Gemma 3 technical report tables are image-embedded (arXiv:2503.19786).

(3) Structured output: **no native function-calling template**. Google released FunctionGemma as a separate variant. Relies on constrained decoding frameworks. The IFEval score of 90.2% suggests the model follows complex instructions well, making grammar-constrained JSON generation likely reliable.

(4) Latency: **estimated ~28–35 tok/s** on Orin Nano (extrapolated from Gemma-2-2B at 34.97 tok/s, scaling for increased model size). NVIDIA blog confirms Gemma-3-4B runs on Orin Nano 8GB. NVIDIA Developer NVIDIA Developer TensorRT-LLM INT4 AWQ: N/A.

(5) Memory at Q4_K_M: ~**2.5 GB**. INT4 AWQ: ~2.0 GB.

(6) Licence: **Gemma Terms of Use**—commercial use permitted with acceptance of Google ToU.

(7) Weaknesses: MMLU base score of 59.51 (lowest among the 4B-class candidates) suggests weaker factual knowledge breadth. No native function calling increases dependence on inference framework for structured output.

(8) Control-loop evaluation: **No.**

Llama-3.2-3B-Instruct — most mature edge deployment ecosystem

(1) 3.2B parameters, dense decoder-only transformer, pruned and distilled from Llama-3.1-8B. NVIDIA Medium Released **September 2024**. Architecture family: Llama 3.2.

(2) Reasoning benchmarks (vendor-reported, Microsoft cross-eval pipeline): MMLU 61.8, GSM8K 75.6, BBH 55.4, huggingface arXiv **IFEval 77.4%** LLM Stats (Meta-reported), LLM Stats ARC-C 76.1, MMLU-Pro 39.2. huggingface arXiv Weakest reasoning of the four top candidates by a clear margin—BBH of 55.4 versus Phi-4-mini's 70.4.

(3) Structured output: tool calling via JSON-based parser in vLLM, but vLLM documentation explicitly warns: **"Llama's smaller models frequently fail to emit tool calls in the correct**

format" vLLM (https://docs.vllm.ai/en/latest/features/tool_calling/). BFCL V2 score: 67.0 (Meta-reported). Medium Independent fine-tuning benchmarks rank it **#1 for tunability**—the model showing the largest gains from task-specific fine-tuning distil labs (distillabs.ai, independent). distil labs

(4) Latency: **43.07 tok/s** decode on Orin Nano Super (NVIDIA official, MLC API q4f16_ft— nvidia the only direct 3B-class benchmark on this hardware). TTFT estimated ~0.6–0.8s. TensorRT-LLM INT4 AWQ: N/A.

(5) Memory at Q4_K_M: ~**2.0 GB**. Comfortably fits with ample headroom.

(6) Licence: **Llama 3.2 Community License**—commercial use permitted with conditions (>700M MAU requires Meta agreement).

(7) Weaknesses: MMLU-Pro of 39.2 is concerning for multi-factor priority reasoning. Tool-calling format failures documented in vLLM. However, constrained decoding eliminates format concerns.

(8) Control-loop evaluation: **No.**

Llama-3.1-8B-Instruct — upper reasoning bound within the SLM range

(1) 8.0B parameters, dense decoder-only transformer. Released **July 2024**. Architecture family: Llama 3.1.

(2) MMLU 68.1, GSM8K 82.4, BBH 63.4, huggingface arXiv **IFEval 80.4%,** LLM Stats LLM Stats ARC-C 83.1 arXiv (all vendor-reported via Microsoft cross-eval). huggingface Groq's Llama 3 8B tool-use checkpoint achieved **90.76% on BFCL** X (independent, ICML 2025).

(3) Structured output: native JSON-based tool calling; more reliable than the 3B variant. BFCL tool-use fine-tune demonstrates strong structured output capability.

(4) Latency: **19.14 tok/s** decode on Orin Nano Super (NVIDIA official). nvidia Memory-constrained: 4.9 GB Q4_K_M leaves <3 GB for KV cache and OS. TTFT estimated ~1.5–2.5s. This model pushes the hardware limits of Orin Nano 8GB.

(5) Memory at Q4_K_M: ~**4.9 GB**. Tight fit on 8GB—functional but context-length constrained.

(6) Licence: **Llama 3.1 Community License**.

(7) Weaknesses: at 8B, decode speed drops to ~19 tok/s (less than half Llama-3.2-3B's throughput). The 4.9 GB memory footprint limits concurrent operations on resource-constrained hardware.

(8) Control-loop evaluation: **No.**

Remaining general-purpose SLMs (abbreviated profiles)

Phi-3.5-mini (3.8B): MMLU 65.5, GSM8K 76.9, BBH 63.1, ARC-C 84.6 arXiv (vendor-reported). huggingface ~38.1 tok/s on Orin Nano (NVIDIA official). nvidia MIT licence. LLM Stats Superseded by Phi-4-mini on every reasoning benchmark except ARC-C (84.6 vs 83.7). No native function calling. Still viable but dominated.

Qwen2.5-3B-Instruct: MMLU 65.0, GSM8K 80.6, BBH 56.2, ARC-C 82.6 arXiv (vendor-reported). huggingface ~2.0 GB at Q4_K_M. **Qwen Research License—restricted commercial use** Qwen (unlike Qwen3-4B's Apache 2.0). Superseded by Qwen3-4B on every metric. Only relevant if Qwen3-4B proves incompatible with a specific deployment constraint.

Gemma-2-2B: MMLU 52.2, GSM8K 24.3, BBH 41.9 Scribd (all base model, vendor-reported). ~35 tok/s on Orin Nano (NVIDIA official). ~1.6 GB at Q4_K_M. nvidia The GSM8K base score of 24.3% is insufficient for numerical reasoning in traffic scenarios. Dominated by every 3B+ model.

Mistral-Nemo (12B): MMLU 68.0 (vendor-reported). ** Mistral AI Not deployable on Orin Nano 8GB**—Q4_K_M weights alone consume ~7.3 GB, leaving <1 GB for OS and KV cache. Included only as an upper-bound reference. Apache 2.0 licence.

SmolLM2-1.7B: MMLU-Pro 19.3, GSM8K 48.2, BBH 32.2, IFEval 56.7 huggingface PromptLayer (vendor-reported). **64.5 tok/s** on Orin Nano (NVIDIA official, fastest model tested). nvidia ~1.1 GB at Q4_K_M. Apache 2.0 licence. Reasoning scores are insufficient for multi-factor priority decisions, but useful as a speed anchor and potential speculative-decoding draft model.

Domain-adapted traffic LLMs

Traffic-R1 (3B) — released weights, full profile

(1) 3B parameters, based on **Qwen2.5-3B**, fine-tuned via reinforcement learning on traffic signal control data. Hugging Face Released **August 2025** (arXiv:2508.02344). arXiv Public checkpoint: "Traffic-R1-3B (Public 0.1)" on HuggingFace (https://huggingface.co/Season998/Traffic-R1).

(2–3) No standard reasoning benchmarks reported (MMLU, GSM8K, etc.). The model claims performance matching GPT-4o on TSC tasks and reports deployment managing signals for **55,000+ daily drivers**, reducing average queue lengths by >5% (vendor-reported). Structured output: produces signal phase selections compatible with LLMLight's CityFlow framework, with noted prompt format incompatibilities. Hugging Face

(4–5) No published edge inference benchmarks. Inherits Qwen2.5-3B's inference profile (~2.0 GB at Q4_K_M, estimated ~40 tok/s on Orin Nano).

(6) Licence: inherits **Qwen Research License** from base model (restricted commercial use).

(7) Weaknesses: early/partial checkpoint (not all training data included due to commercial/privacy concerns); evaluated on CityFlow, **not SUMO**; being upgraded to Qwen3/VL (current weights may become outdated). No general reasoning benchmarks means its capacity for novel priority scenarios outside training distribution is unknown.

(8) Control-loop evaluation: **Yes**—designed for and reportedly deployed in real-world TSC.

LightGPT (Lai et al., KDD 2025) — released weights, full profile

(1) Original: **13B parameters**, Llama-2-13B base, Hugging Face fine-tuned via imitation learning from GPT-4 traces + RL policy refinement. GitHub arXiv Expanded family includes Qwen2/Llama3 variants. GitHub Published at **KDD 2025** (peer-reviewed). GitHub: https://github.com/usail-hkust/LLMTSCS.

(2–3) No standard reasoning benchmarks. Outperforms GPT-4 on 10 real-world TSC datasets (ATT, AQL, AWT metrics). arXiv Uses Chain-of-Thought reasoning for phase decisions. GitHub arXiv 15 human expert evaluations confirm interpretability ADS (peer-reviewed, KDD 2025). arXiv

(4–5) 13B original is too large for Orin Nano 8GB. Smaller Qwen2/Llama3 variants would fit but lack published benchmarks.

(6) Licence: **MIT** (model card), though base model licences (Llama-2, Qwen2) add constraints.

(7) Each intersection agent runs independently—no inter-intersection cooperation. Training depends on GPT-4 distillation. Evaluated on CityFlow, **not SUMO**.

(8) Control-loop evaluation: **Yes**—operates as a direct TSC control agent in CityFlow simulation.

Method-only benchmark anchors (four-field summaries)

CoLLMLight (Yuan et al., 2025, arXiv:2503.11739): First cooperative LLM agent framework for network-wide TSC. Constructs spatiotemporal graphs capturing inter-intersection dynamics arXiv with asynchronous cooperative reasoning and complexity-aware depth adaptation. OpenReview Wraps **Llama-3.1 (8B/70B), Qwen3 (8B/32B), DeepSeek-R1 distilled models**. The Moonlight OpenReview Outperforms LLMLight and VLMLight on CityFlow (ATT, AWT, AQL). Code public on GitHub; no pre-trained weights released—users must fine-tune on their own base LLM.

LA-Light (Wang et al., 2024, arXiv:2403.08337): Hybrid LLM-RL framework with five-stage decision process where the LLM handles anomalous/long-tail events while RL handles routine control. arXiv arXiv Wraps **GPT-4 via API**. Evaluated on **SUMO** (via TransSimHub)—the only surveyed method with direct SUMO evaluation. Code public; **closed API dependency**.

TrafficGPT (Zhang et al., 2024, Transport Policy vol. 150): Orchestration framework coupling LLMs with Traffic Foundation Models for task decomposition and decision support. arXiv Wraps **ChatGPT (GPT-3.5/4)**. Not a direct TSC control agent—more of a traffic analytics tool. Demo code public; **closed API dependency**.

VLMLight (Wang et al., 2025, arXiv:2505.19486): Dual-branch architecture integrating vision-language models for safety-critical TSC. A meta-controller selects between fast RL and deliberative multi-agent LLM reasoning for emergency scenarios. arXiv Reduces **emergency vehicle waiting time by up to 65%.** arXiv Uses custom image-based simulator (not SUMO or CityFlow). Code planned for public release; likely depends on closed VLM/LLM APIs.

Additional 2025–2026 systems discovered: CuraLight (arXiv:2604.05663, April 2026) uses RL-assisted LLM fine-tuning via debate-guided data curation; SignalClaw (arXiv:2604.05535, April 2026) uses LLMs as mutation operators for evolutionary program synthesis of interpretable control skills; arXiv iLLM-TSC (arXiv:2407.06025) integrates RL with LLM correction on SUMO via TransSimHub (GPT-4 dependent). arXiv None release weights.

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§3. Comparison table

Runnable general-purpose models

Model	Params	MMLU	GSM8K	BBH	IFEval	ARC-C	Decode (tok/s) Orin Nano	Mem Q4_K_M (GB)	Licence	JSC
Qwen3-4B	4.0B	73.0†	87.8†	72.6†	87.8‡ / 68.9§	N/R	~32–38 est.	~2.5	Apache 2.0	Nati FC,
Phi-4-mini	3.8B	67.3	88.6	70.4	~63.0	83.7	~35–40 est.	~2.3	MIT	Nati FC,
Gemma-3-4B-IT	3.88B	~59.5††	89.2	N/F	**90.2**	N/F	~28–35 est.	~2.5	Gemma ToU	No FC
Llama-3.2-3B	3.2B	61.8	75.6	55.4	77.4	76.1	**43.1**	~2.0	Llama CL	Wea
Llama-3.1-8B	8.0B	68.1	82.4	63.4	80.4	83.1	19.1	~4.9	Llama CL	Nati
Phi-3.5-mini	3.8B	65.5	76.9	63.1	N/F	84.6	38.1	~2.3	MIT	No FC
Qwen2.5-3B	3.1B	65.0	80.6	56.2	N/F	82.6	~40–45 est.	~2.0	Qwen RL	No FC

Model	Params	MMLU	GSM8K	BBH	IFEval	ARC-C	Decode (tok/s) Orin Nano	Mem Q4_K_M (GB)	Licence	JSC
Gemma-2-2B	2.6B	52.2	24.3††	41.9††	N/F	55.7††	35.0	~1.6	Gemma ToU	No
SmolLM2-1.7B	1.7B	19.3¶	48.2	32.2	56.7	~51.7	**64.5**	~1.1	Apache 2.0	No
Mistral-Nemo	12B	68.0	N/F	N/F	N/F	N/F	✗ OOM	~7.3	Apache 2.0	Strc (her

All benchmarks vendor-reported unless noted. † = base model scores (instruct higher). ‡ = thinking mode. § = non-thinking mode. †† = base model only. ¶ = MMLU-Pro, not standard MMLU. N/F = not found. N/R = not reported. Decode tok/s from NVIDIA official MLC API benchmarks where available; "est." = extrapolated. "Llama CL" = Llama Community License. "Qwen RL" = Qwen Research License (restricted). "Gemma ToU" = Gemma Terms of Use.

Runnable domain-adapted models

Model	Params	Base	TSC metrics reported	Simulator	Weights	Licence
Traffic-R1	3B	Qwen2.5-3B	ATT, AQL (matches GPT-4o, vendor)	CityFlow	✅ HuggingFace	Qwen RL
LightGPT	13B	Llama-2-13B	ATT, AQL, AWT (KDD 2025, peer-reviewed)	CityFlow	✅ HuggingFace	MIT

Method-only benchmark anchors (not deployable candidates)

System	Base LLM	Simulator	Weights?	API dependency
CoLLMLight	Llama-3.1/Qwen3/DeepSeek-R1	CityFlow	Code only	Open-source LLMs
LA-Light	GPT-4	**SUMO**	Code only	✗ Closed API
TrafficGPT	ChatGPT	Multi-modal	Demo only	✗ Closed API
VLMLight	VLM + LLM	Custom	Planned	✗ Likely closed
iLLM-TSC	GPT-4	**SUMO**	Code only	✗ Closed API

§4. Decision matrix across three scenarios

(a) Maximum reasoning quality, latency secondary

Qwen3-4B in thinking mode is the clear leader. Its MATH-500 of 95.16%, AIME'24 of 70.0%, and IFEval strict of 87.75% (all vendor-reported) dramatically outperform every other sub-8B model. Readthedocs The thinking mode generates extended reasoning chains before answering, adding seconds of latency— BuildMVPFast acceptable only when simulation time-steps are long or reasoning occurs offline (e.g., pre-computing priority policies for scenario libraries).

Llama-3.1-8B provides the highest-quality single-pass reasoning in this survey (MMLU 68.1, IFEval 80.4) without thinking-mode complexity, at the cost of ~19 tok/s decode and tight memory on Orin Nano. For a simulation-only context with no real hardware constraint, it remains viable.

Phi-4-mini is the strongest reasoner among sub-4B models in single-pass mode (BBH 70.4, MMLU-Pro 52.8), and avoids the thinking/non-thinking mode complexity entirely.

Ranking: Qwen3-4B (thinking) > Llama-3.1-8B > Phi-4-mini > Qwen3-4B (non-thinking) ≈ Gemma-3-4B-IT.

(b) Best latency/reasoning trade-off for a 500 ms simulation decision budget

The 500 ms budget includes prompt assembly, inference (TTFT + decode for ~50-token JSON output), and output parsing. On actual Jetson Orin Nano hardware, **no 3–4B model achieves this budget.** Even with prefix caching (reducing TTFT from ~600 ms to ~50–125 ms by caching the system prompt), 50-token decode at 18–38 tok/s requires 1.3–2.8 seconds. The total ranges from ~1.4 s (best case, SmolLM2 at 64.5 tok/s with aggressive caching) to ~3.5 s (Gemma-3-4B without caching).

For **simulation-first research** where inference runs on a desktop GPU (e.g., RTX 4060/4090) rather than Jetson hardware, the budget becomes achievable. An RTX 4090 delivers **~100–200 tok/s** for 3–4B Q4 models via llama.cpp, placing total inference (with prefix caching) at **~150–400 ms**—within budget.

Under these simulation conditions, the models that best balance reasoning quality against TTFT-dominated latency on short outputs are:

- **Phi-4-mini**: single-pass with no thinking overhead, strong BBH (70.4), predictable latency. No risk of runaway reasoning tokens.

- **Qwen3-4B (non-thinking, hard switch)**: comparable quality to Phi-4-mini in non-thinking mode, but the documented risk of hidden reasoning tokens (ollama/ollama#12907) introduces latency unpredictability. **Mitigation**: use `enable_thinking=False` via chat template (not the soft `/no_think` switch), and set a max-token limit on generation.
- **Llama-3.2-3B**: fastest official Orin Nano throughput (43.1 tok/s), lowest memory footprint among 3B+ models, but weakest reasoning (BBH 55.4).

Speculative decoding using SmolLM2-1.7B as a draft model for Phi-4-mini or Qwen3-4B could improve decode throughput by **1.5–2×** on GPU hardware (demonstrated on Jetson AGX Orin, NVIDIA Developer Forums), but adds implementation complexity and memory pressure. Prefix caching is supported natively in llama-server (the `cache_prompt` parameter, enabled by default) and represents the lowest-complexity latency mitigation.

Ranking for this scenario: Phi-4-mini (most predictable) > Qwen3-4B non-thinking (higher ceiling, less predictable) > Llama-3.2-3B (fastest but weakest reasoning).

(c) Most parameter-efficient for simulation-first dissertation research

For a researcher who will never deploy real hardware—running inference on a desktop GPU with SUMO in the loop—the binding constraints shift to **reasoning quality per parameter, structured output reliability, tooling maturity, and licence freedom**.

Qwen3-4B offers the best reasoning-per-parameter ratio. Its base model matches Qwen2.5-7B (nearly 2× larger) on MMLU, BBH, and GSM8K. Apache 2.0 licensing removes legal friction. NVIDIA's Jetson tutorial validates the model for future speculative-framing sections of the dissertation. The Qwen3-4B-Instruct-2507 update specifically improves tool use and instruction following.

Phi-4-mini at 3.8B provides comparable non-thinking reasoning with MIT licensing and official ONNX support from Microsoft—useful if the dissertation explores multiple inference frameworks. Its lack of thinking mode is actually an advantage for reproducibility: behaviour is deterministic given the same prompt and sampling parameters.

Gemma-3-4B-IT merits consideration for its **IFEval 90.2%**—the highest instruction-following score in this survey. For a system that parses model output as JSON for priority decisions, instruction adherence directly predicts output parseability. However, its lower MMLU base score (59.5) suggests narrower factual coverage.

Traffic-R1 (3B, Qwen2.5-3B base) serves as a valuable domain-specific benchmark anchor despite being evaluated on CityFlow rather than SUMO. Comparing a general-purpose SLM against Traffic-R1 on overlapping metrics would strengthen the dissertation's contribution.

Ranking: Qwen3-4B (best quality/param) > Phi-4-mini (simplest, most reproducible) > Gemma-3-4B-IT (best structured output adherence) > Traffic-R1 (domain anchor).

§5. Gaps, caveats, and unverifiable claims

No model in this survey has been evaluated for semantic traffic priority reasoning. All reasoning capability assessments are proxied through general benchmarks (MMLU, BBH, GSM8K). The assumption that strong general reasoning transfers to traffic priority decisions is plausible but untested. Traffic-R1 and LLMLight operate as phase-selection controllers—a narrower task than the semantic priority reasoning described in the research question.

Jetson Orin Nano TTFT data is sparse and inconsistent. Eric Liu's measurements (community-reported) provide the only published TTFT values on Orin Nano, and only for sub-2B models. All TTFT estimates for 3–4B models are extrapolated. NVIDIA's official benchmarks report only decode throughput, not TTFT. The MLC API benchmarks use q4f16_ft quantization, not llama.cpp Q4_K_M—these are reasonable proxies (community tests confirm similar throughput) but not identical.

TensorRT-LLM INT4 AWQ benchmarks do not exist for Orin Nano. NVIDIA's TensorRT Edge-LLM tutorial confirms Qwen3-4B runs on Orin Nano with INT4 AWQ but publishes no performance numbers. Every cell in the TensorRT column is "N/A."

Quantisation impact on reasoning is unquantified for most models. All benchmark scores are FP16/BF16. Q4_K_M quantisation typically degrades scores by 1–5%, with reasoning tasks affected more than factual recall (community consensus, no peer-reviewed study found specific to these models).

Qwen3-4B's thinking mode reliability is uncertain. GitHub issues document the model emitting reasoning tokens even when thinking is nominally disabled. For a control system requiring predictable latency, this is a non-trivial engineering risk that requires the hard `enable_thinking=False` switch and output-length caps.

Structured output reliability is model-agnostic when using constrained decoding. The JSONSchemaBench study (arXiv:2501.10868, independent) demonstrates 95–99% JSON conformance rates with grammar-constrained decoding (llama.cpp GBNF, XGrammar). **This means structured output reliability is primarily an infrastructure choice, not a model differentiator**, as long as constrained decoding is employed. The native function-calling scores reported above (Qwen3-4B at 0.920, Phi-4-mini at 0.780) reflect unconstrained generation quality and represent a floor, not a ceiling.

Domain-adapted models use CityFlow, not SUMO. Traffic-R1, LLMLight, and CoLLMLight all evaluate on CityFlow. Only LA-Light and iLLM-TSC use SUMO, and both depend on closed GPT-4 APIs. No open-weight traffic-domain model has been evaluated on SUMO. Bridging this gap—evaluating a general-purpose SLM on SUMO for priority reasoning—would be a direct contribution of the dissertation.

Jailbreak and safety evaluations are aggregate, not model-specific. The finding that 47.6% of SLMs show high jailbreak susceptibility (arXiv:2503.06519) is a population-level statistic. Individual ASR scores for the specific models in this survey were not extractable from available materials. No SLM in this survey has undergone safety evaluation in a traffic control context.

The 500 ms decision budget is infeasible on Orin Nano for any 3B+ model. Even with prefix caching, decode latency for 50 tokens exceeds the budget by 3–5×. The budget is achievable only on desktop GPUs (RTX 4060/4090) or with models under 1B parameters generating fewer than 15–20 tokens—neither of which meets the semantic reasoning requirement. This constraint is relevant only for the speculative hardware-framing sections of the dissertation; simulation-time performance on a desktop GPU is the practical binding constraint.

Conclusion

The landscape of sub-8B SLMs for traffic priority reasoning presents three distinct frontiers rather than a single winner. **Qwen3-4B** offers the highest reasoning ceiling of any sub-5B model, with native function calling and Apache 2.0 licensing, but introduces thinking-mode latency risk that requires engineering discipline to manage. **Phi-4-mini** provides the most predictable inference behaviour and strongest single-pass compositional reasoning (BBH 70.4), making it the safest choice for a system requiring deterministic latency. **Gemma-3-4B-IT's** IFEval score of 90.2% makes it uniquely suited for parsing-critical applications, though its weaker knowledge breadth limits its utility for novel scenarios.

The most impactful dissertation contribution would combine a general-purpose SLM (Qwen3-4B or Phi-4-mini) with grammar-constrained decoding in SUMO—a configuration no published work has evaluated. Benchmarking this against Traffic-R1's CityFlow results would provide the first cross-simulator, cross-architecture comparison in the LLM-for-TSC literature. The structured output reliability question is, in practice, solved at the infrastructure layer by constrained decoding rather than by model selection—a finding that should reshape how the dissertation frames its model-choice contribution.